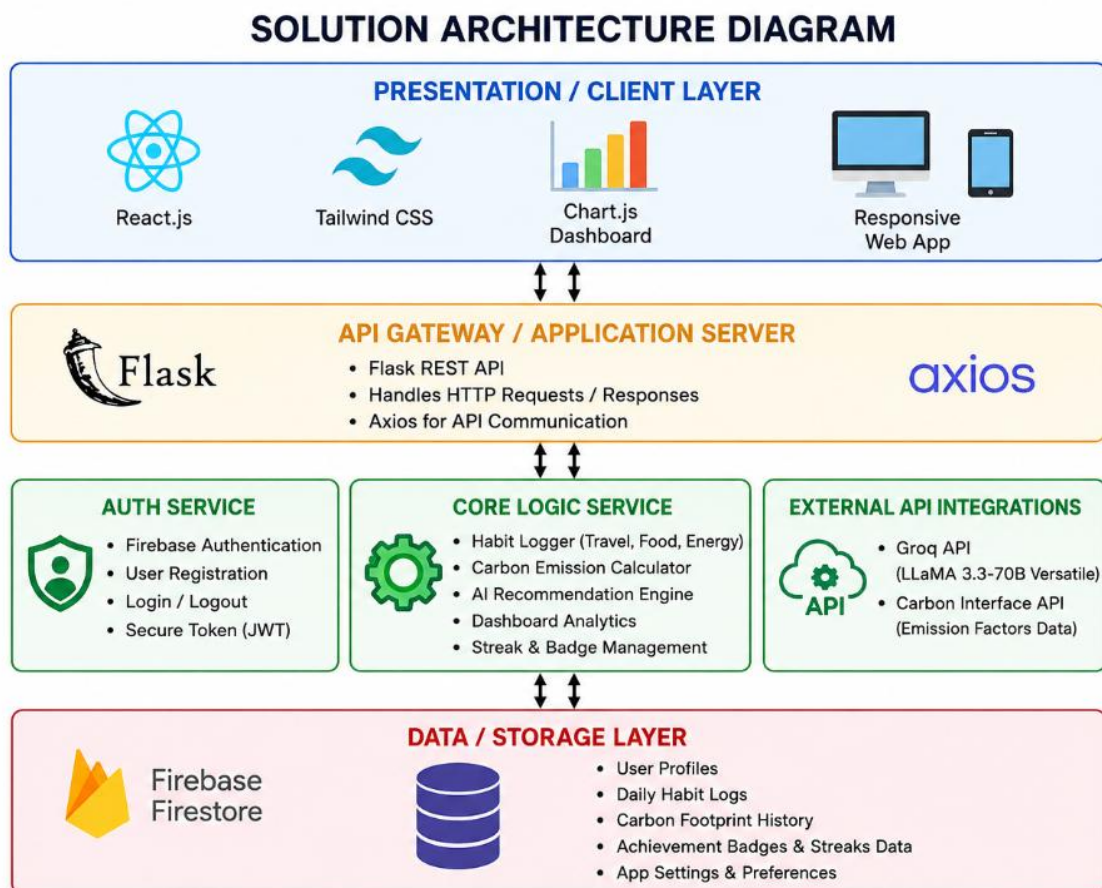


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	EcoTrack – AI-Powered Carbon Footprint Tracker
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

COMPONENT DESCRIPTION TABLE		
Component Name	Description / Role in Architecture	Technologies Used
 PRESENTATION LAYER (Client Layer)	Provides the user interface for interacting with EcoTrack. Users can log daily habits (travel, food, energy), view carbon footprint dashboard, analytics charts, AI suggestions, badges, and reports.	    React.js Tailwind CSS Chart.js JavaScript
 API GATEWAY (Application Server)	Acts as the communication bridge between the frontend and backend services. Handles API requests, authentication validation, routing, and responses.	  Flask (REST API) Axios (API Communication)
 CORE SERVICES (Microservices / Business Logic)	Contains the core business logic and services that power EcoTrack including authentication, carbon calculation, AI recommendations, dashboard analytics, streak and badge management, and external API integrations.	    Firebase Auth (Auth Service) Python + Flask (Core Logic) Groq API (AI Suggestions) Carbon Interface API (Emission Factors)
 DATA / STORAGE LAYER (Database Layer)	Stores all application data securely including user profiles, habit logs, carbon footprint history, achievement badges, streaks, and app settings.	  Firebase Firestore (Cloud NoSQL Database) Firebase Authentication (User Management)